

Ziyu (Rick) Xu

Robotics Institute, Carnegie Mellon University

Research Interests: Robot Foundation Models, VLA Adaptation, and Real-Robot Learning

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Education

Carnegie Mellon University (CMU)

Master of Science in Robotics (MSR)

Aug. 2026 – Expected 2028

Pittsburgh, PA

University of Michigan, Ann Arbor (UM)

B.S.E. in Mechanical Engineering (Robotics Track); Minor in Computer Science GPA: 3.85/4.0 (Top 5%)

Oct. 2024 – May 2026

- **Core Coursework:** Robot Learning, RL, Embedded Systems, Optimal Control, Convex Optimization

Shanghai Jiao Tong University (SJTU)

B.S.E. in Electrical and Computer Engineering

Sep. 2022 – Jul. 2026

GPA: 3.56/4.0 (Top 20%)

- **Core Coursework:** Algorithms, Mathematical Foundations, Probability Theory, Signals and Systems

Research Experiences and Internships

Unitree Robotics, R&D Department, Embodied AI Team

May 2026 – Present

Research Intern, Humanoid Robot Foundation Models

Hangzhou, China

- **Real-Robot Policy Infrastructure:** Built and validated a closed-loop stack on Unitree H2/G1 humanoids integrating teleoperation, VLA/WAM and whole-body policy adapters, rollout logging, evaluation, controller interfaces, and policy fine-tuning; enabled one-command launches across platforms.
- **VLA Adaptation and Post-Training:** Deployed and fine-tuned Isaac GR00T/OpenPI policies with SONIC whole-body control using real-robot rollout traces. Developing phase-aware adaptation across embodiments by aligning semantic task progress and execution timing.
- **Reproducible Evaluation:** Implemented an 8-task, 24-scene manipulation benchmark with background variation. Built PICO-based scene digitization and AR-guided reconstruction with reusable layouts, achieving 2.5 cm median placement error.

Intelligent Control Lab, Carnegie Mellon University

Aug. 2026 – Present

Graduate Student Researcher, advised by Prof. Changliu Liu

Pittsburgh, PA

- **Whole-Body and Safe Locomotion Research (ongoing):** Investigating whole-body control, safety with control barrier functions (CBFs), and world-model-based locomotion.
- **Dexmate Bimanual Manipulation Demo, IROS 2026:** Contributed to a bimanual block-stacking demonstration combining agent planning with policy execution. Implemented the agent component, prepared the on-site environment, adapted the system to different grippers, and performed system calibration.

Shanghai Artificial Intelligence Lab & IRMV, SJTU

Apr. 2026 – Present

Research Assistant, Directed by Prof. Junchi Yan and Prof. Hesheng Wang

Shanghai, China

- **Iterative Visual KV Compression:** Studied errors from reusing retained visual KV entries after neighboring tokens are removed in multi-turn and streaming inference. Developed context-aware re-encoding at original positions and compared with inheritance and restoration under matched memory and runtime budgets.

Ziyu (Rick) Xu

Computational Autonomy and Robotics Lab, UM

Research Assistant, Directed by Prof. Maani Ghaffari

Research experience continued

Jun. 2025 – Apr. 2026

Ann Arbor, MI

- **Learning-Augmented Lie-MPC:** Developed and field-tested an autonomous surface vehicle controller that estimates residual wind/current disturbances online and corrects the dynamics used by MPC. Achieved a 20% improvement in trajectory-tracking performance over baseline MPC in real-ASV experiments.

Intelligent Robotics and Autonomy Lab , UM

Research Assistant, Directed by Prof. Vasileios Tzoumas

Oct. 2024 – Apr. 2026

Ann Arbor, MI

- **Multi-Robot Simulation and ROS2:** Rebuilt a Unity + AirSim multi-drone testbed with revised dynamics and sensing; migrated SlideSLAM communication from ROS1 to ROS2 for bandwidth-limited active-SLAM evaluation.

Publications and Manuscripts

- Xiaohan Wang^{*}, **Ziyu Xu^{*}**, Xuyi Yang. “Re-encoding What You Keep After Every Compression: Retained Visual KV Is Computed in a Context That No Longer Exists”, *Submitted to ICLR 2027*. ^{*}*Equal contribution*.
- Xuyi Yang, Xiaohan Wang, Wenhao Zhang, and **Ziyu Xu**. “StreamJIT: Continuous Observation and Just-in-Time Deliberation for Proactive Streaming Video Assistants”, *Submitted to ICLR 2027*.
- Yinan Dong, **Ziyu Xu**, Tsimafei Lazouski, Sangli Teng, and Maani Ghaffari. “Online Learning-Enhanced Lie Algebraic MPC for Robust Autonomous Surface Vehicle Control”, *IEEE Robotics and Automation Letters (RA-L)*, revised manuscript under review, 2025.
- C. Yuan, J. Jiang, K. Yang, **Z. Xu**, . . . , and Y. Ban. “Systematic Evaluation and Guidelines for Segment Anything Model in Surgical Video Analysis”, *npj Digital Surgery*, accepted, 2024.

Selected Awards

Gold Medal, University Physics Competition (2023); Dean’s Honor List, University of Michigan (2024, 2025).

Skills

- **Robot Learning:** VLA post-training, real-robot deployment, whole-body policy integration, RL, teleoperation, rollout evaluation
- **ML and Systems:** PyTorch, Python/C++ , ROS/ROS2, Linux, Docker, Git, OpenCV, visual KV compression